

Chapter 1

Primal and dual problems

In this chapter we will start with generalities on the transport problem from a measure μ on a space X to another measure ν on a space Y . In general, X and Y can be complete and separable metric spaces, but soon we will focus on the case where they are the same subset $\Omega \subset \mathbb{R}^d$ (often compact). The cost function $c : X \times Y \rightarrow [0, +\infty]$ will be possibly supposed to be continuous or semi-continuous, and then we will analyze particular cases (such as $c(x, y) = h(x - y)$ for strictly convex h).

For the sake of the exposition, the structure of this chapter is somehow involved, and deserves being explained. In Section 1.1 we present the problems by Monge and Kantorovich and prove existence for the Kantorovich Problem (KP). In Section 1.2 we present the dual problem (DP), but we do not prove the duality result $\min(\text{KP}) = \sup(\text{DP})$. In Section 1.3, taking this duality result as proven, we discuss cases where the solution of (KP) turns out to be induced by a transport map, hence solving (MP). Sections 1.4 and 1.5 are devoted to counter-examples to the existence for (MP) and to the equality $\min(\text{MP}) = \min(\text{KP})$. In Section 1.6, we introduce the notion of cyclical monotonicity, which allows to prove the duality $\min(\text{KP}) = \sup(\text{DP})$, as well as stability results and sufficient optimality conditions. We also give an independent proof of duality, based on notion of convex analysis that we actually introduce in a Memo Box in the same section 1.6. The chapter is concluded by a long discussion section, 1.7.

1.1 Kantorovich and Monge problems

The starting point of Optimal Transport is the classical problem by Monge ([239]) which reads in its most general version, and in modern language, as follows.

Problem 1.1. Given two probability measures $\mu \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Y)$ and a cost function $c : X \times Y \rightarrow [0, +\infty]$, solve

$$(\text{MP}) \quad \inf \left\{ M(T) := \int c(x, T(x)) d\mu(x) : T_{\#}\mu = \nu \right\}, \quad (1.1)$$

where we recall that the measure denoted by $T_{\#}\mu$ is defined through $(T_{\#}\mu)(A) := \mu(T^{-1}(A))$ for every A , and is called *image measure* or *push-forward* of μ through T .

As we pointed out in the introduction, Problem (MP)¹ is difficult because of its constraint. In particular, this constraint on T is not closed under weak convergence: see **Ex(1)** or consider $T_n(x) = \sin(nx)$ on $[0, 2\pi]$.

Because of this difficulty, we will forget (MP) for a while and pass to the generalization that appears as natural from the work of Kantorovich ([201]):

Problem 1.2. Given $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$ and $c : X \times Y \rightarrow [0, +\infty]$, we consider the problem

$$(KP) \quad \inf \left\{ K(\gamma) := \int_{X \times Y} c \, d\gamma : \gamma \in \Pi(\mu, \nu) \right\}, \quad (1.2)$$

where $\Pi(\mu, \nu)$ is the set of the so-called *transport plans*, i.e.

$$\Pi(\mu, \nu) = \{ \gamma \in \mathcal{P}(X \times Y) : (\pi_x)_\# \gamma = \mu, (\pi_y)_\# \gamma = \nu \}$$

where π_x and π_y are the two projections of $X \times Y$ onto X and Y , respectively. These probability measures over $X \times Y$ are an alternative way to describe the displacement of the particles of μ : instead of specifying, for each x , which is the destination $T(x)$ of the particle originally located at x , we specify for each pair (x, y) how many particles go from x to y . More precisely, the value $\gamma(A \times B)$ denotes the amount of mass moving from A to B . It is clear that this description allows for more general movements, since from a single point x particles can a priori move to different destinations y . If multiple destinations really occur, then this movement cannot be described through a map T . Note that the constraints on $(\pi_x)_\# \gamma$ and $(\pi_y)_\# \gamma$ exactly mean that we restrict our attention to the movements that really take particles distributed according to the distribution μ and move them onto the distribution ν .

The minimizers for this problem are called *optimal transport plans* between μ and ν . Should γ be of the form $(\text{id}, T)_\# \mu$ for a measurable map $T : X \rightarrow Y$ (i.e. when no splitting of the mass occurs), the map T would be called *optimal transport map* from μ to ν .

Remark 1.3. It can be easily checked that $(\text{id}, T)_\# \mu$ (this transport plan will be denoted by γ_T) belongs to $\Pi(\mu, \nu)$ if and only if T pushes μ onto ν (i.e. $\nu(A) = \mu(T^{-1}(A))$ for any Borel set A) and the functional takes the form $\int c(x, T(x)) d\mu(x)$, thus generalizing the Monge Problem.

This generalized problem by Kantorovich is much easier to handle than the original one proposed by Monge: for instance in the Monge case we would need existence of at least a map T satisfying the constraints. This is not verified when $\mu = \delta_0$, if ν is not a single Dirac mass (see Section 1.4). On the contrary, there always exist transport plans in $\Pi(\mu, \nu)$ (for instance $\mu \otimes \nu \in \Pi(\mu, \nu)$). Moreover, one can state that (KP) is the relaxation of the original problem by Monge (something which will be made precise in Section 1.5, and it means, roughly speaking, that (KP) is somehow the minimal extension of (MP) which has some chances to admit a minimizer).

Anyway, it is important to note that an easy use of the Direct Method of Calculus of Variations proves that a minimum does exist. This means that we take a minimizing sequence, we say that it is compact in some topology (here it is the weak convergence

¹To clarify the notation, let us stress that we use (MP) to denote the name of the minimization problem, as well as (KP), (BP) and many others later on. For its minimal value, we write $\min(\text{MP})$.

of probability measures), we find a limit, and prove semi-continuity (or continuity) of the functional we minimize, so that the limit is a minimizer.

Box 1.1. – Memo – Weierstrass criterion for the existence of minimizers, semi-continuity

The most common way to prove that a function admits a minimizer is called “direct method in calculus of variations”. It simply consists in the classic Weierstrass Theorem, possibly replacing continuity with semi-continuity.

Definition - On a metric space X , a function $f : X \rightarrow \mathbb{R} \cup \{+\infty\}$ is said to be lower-semi-continuous (l.s.c. in short) if for every sequence $x_n \rightarrow x$ we have $f(x) \leq \liminf_n f(x_n)$.

Definition - A metric space X is said to be compact if from any sequence x_n we can extract a converging subsequence $x_{n_k} \rightarrow x \in X$.

Theorem (Weierstrass) - If $f : X \rightarrow \mathbb{R} \cup \{+\infty\}$ is lower semi-continuous and X is compact, then there exists $\bar{x} \in X$ such that $f(\bar{x}) = \min\{f(x) : x \in X\}$.

Proof - Define $\ell := \inf\{f(x) : x \in X\} \in \mathbb{R} \cup \{-\infty\}$ ($\ell = +\infty$ only if f is identically $+\infty$, but in this case any point in X minimizes f). By definition there exists a minimizing sequence x_n , i.e. points in X such that $f(x_n) \rightarrow \ell$. By compactness we can assume $x_n \rightarrow \bar{x}$. By lower semi-continuity, we have $f(\bar{x}) \leq \liminf_n f(x_n) = \ell$. On the other hand, we have $f(\bar{x}) \geq \ell$ since ℓ is the infimum. This proves $\ell = f(\bar{x}) \in \mathbb{R}$ and this value is the minimum of f , realized at $\bar{x}$.

Box 1.2. – Memo – Weak compactness in dual spaces

Definition - A sequence x_n in a Banach space $\mathcal{X}$ is said to be weakly converging to x , and we write $x_n \rightharpoonup x$, if for every $\xi \in \mathcal{X}'$ (where $\mathcal{X}'$ is the topological dual of $\mathcal{X}$ and $\langle \cdot, \cdot \rangle$ stands for the duality product between these spaces) we have $\langle \xi, x_n \rangle \rightarrow \langle \xi, x \rangle$. A sequence $\xi_n \in \mathcal{X}'$ is said to be weakly-* converging to ξ , and we write $\xi_n \xrightarrow{*} \xi$, if for every $x \in \mathcal{X}$ we have $\langle \xi_n, x \rangle \rightarrow \langle \xi, x \rangle$.

Theorem (Banach-Alaoglu) - If $\mathcal{X}$ is separable and ξ_n is a bounded sequence in $\mathcal{X}'$ then there exists a subsequence ξ_{n_k} weakly converging to some $\xi \in \mathcal{X}'$.

We refer for instance to [91] for all the details on functional analysis.

Box 1.3. – Memo – Duality between C_0 and $\mathcal{M}$

Definition - A finite signed measure λ on a metric space X is a map associating to every Borel subset $A \subset X$ a value $\lambda(A) \in \mathbb{R}$ (we will see in Chapter 4 the case of vector measures, where λ is valued in $\mathbb{R}^d$) such that, for every countable disjoint union $A = \bigcup_i A_i$ (with $A_i \cap A_j = \emptyset$ for $i \neq j$), we have

$$\sum_i |\lambda(A_i)| < +\infty \quad \text{and} \quad \lambda(A) = \sum_i \lambda(A_i).$$

We denote by $\mathcal{M}(X)$ the set of finite signed measures on X . To such measures we can associate a positive scalar measure $|\lambda| \in \mathcal{M}_+(X)$ through

$$|\lambda|(A) := \sup \left\{ \sum_i |\lambda(A_i)| : A = \bigcup_i A_i \text{ with } A_i \cap A_j = \emptyset \text{ for } i \neq j \right\}.$$

Theorem (Riesz representation theorem) - Suppose that X is a separable and locally compact metric space. Let $\mathcal{X} = C_0(X)$ be the space of continuous function on X vanishing at infinity, i.e. $f \in C_0(X) \iff f \in C(X)$ and for every $\varepsilon > 0$ there exists a compact subset $K \subset X$ such that $|f| < \varepsilon$ on $X \setminus K$. Let us endow this space with the sup norm since $C_0(X) \subset C_b(X)$ (this last space being the space of bounded continuous functions on X). Note that $C_0(X)$ is a Banach space, and that it is a closed subset $C_b(X)$. Then every element of $\mathcal{X}'$ is represented in a unique way as an element of $\mathcal{M}(X)$: for all $\xi \in \mathcal{X}'$ there exists a unique $\lambda \in \mathcal{M}(X)$ such that $\langle \xi, \phi \rangle = \int \phi d\lambda$ for every $\phi \in \mathcal{X}$; moreover, $\mathcal{X}'$ is isomorphic to $\mathcal{M}(X)$ endowed with the norm $\|\lambda\| := |\lambda|(X)$.

For signed measures of $\mathcal{M}(X)$ we should call weak-* convergence the convergence in the duality with $C_0(X)$. Yet, another interesting notion of convergence is that in duality with $C_b(X)$. We will call it (by abuse of notation) weak convergence and denote it through the symbol $\rightharpoonup$: $\mu_n \rightharpoonup \mu$ if and only if for every $\phi \in C_b(X)$ we have $\int \phi d\mu_n \rightarrow \int \phi d\mu$ (note that, taking $\phi = 1$, we also have $\mu_n(X) \rightarrow \mu(X)$, which is not the case for the $\overset{*}{\rightharpoonup}$ convergence). Note that $C_0(X) = C_b(X) = C(X)$ if X is compact, and in this case the two notions of convergence are the same. On the other hand, for non-compact X the space $\mathcal{M}(X)$ is not the dual of $C_b(X)$: by Hahn-Banach's theorem, it is possible to produce (see for instance Section 1.3 in [292]) elements of $C_b(X)'$ that only look at the behavior of functions of $C_b(X)$ “out of compact subsets” (i.e. at infinity, or on the boundary). The notion of weak convergence in duality with C_b is also sometimes called *narrow convergence*. For all details about measure theory, we refer for instance to [268].

Box 1.4. – Memo – Weak convergence of probability measures

Probability measures are particular measures in $\mathcal{M}(X)$: $\mu \in \mathcal{P}(X) \iff \mu \in \mathcal{M}_+(X)$ and $\mu(X) = 1$ (note that for positive measures μ and $|\mu|$ coincide).

Definition - A sequence μ_n of probability measures over X is said to be tight if for every $\varepsilon > 0$ there exists a compact subset $K \subset X$ such that $\mu_n(X \setminus K) < \varepsilon$ for every n .

Theorem (Prokhorov) - Suppose that μ_n is a tight sequence of probability measures over a complete and separable metric space X (these spaces are also called *Polish spaces*). Then there exists $\mu \in \mathcal{P}(X)$ and a subsequence μ_{n_k} such that $\mu_{n_k} \rightharpoonup \mu$ (in duality with $C_b(X)$). Conversely, every sequence $\mu_n \rightharpoonup \mu$ is necessarily tight.

Sketch of proof (of the direct implication) - For every compact $K \subset X$, the measures $(\mu_n)|_K$ admit a converging subsequence (in duality with $C(K)$). From tightness, we have an increasing sequence of compact sets K_i such that $\mu_n(K_i^c) < \varepsilon_i = 1/i$ for every i and every n . By a diagonal argument, it is possible to extract a subsequence μ_{n_h} such that $(\mu_{n_h})|_{K_i} \rightharpoonup \nu_i$ (weak convergence as $n \rightarrow \infty$ in duality with $C(K_i)$). The measures ν_i are increasing in i , and define a measure $\mu = \sup_i \nu_i$ (i.e. $\mu(A) = \sup_i \nu_i(A \cap K_i)$). In order to prove $\mu_{n_h} \rightharpoonup \mu$, take $\phi \in C_b(X)$ and write $\int_X \phi d(\mu_{n_h} - \mu) \leq 2\varepsilon_i + \int_{K_i} \phi d(\mu_{n_h} - \nu_i)$. This allows to prove the convergence. Proving $\mu \in \mathcal{P}(X)$ we only need to check $\mu(X) = 1$, by testing with $\phi = 1$.

We are now ready to state some existence results.

Theorem 1.4. *Let X and Y be compact metric spaces, $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$ and $c : X \times Y \rightarrow \mathbb{R}$ a continuous function. Then (KP) admits a solution.*

Proof. We just need to show that the set $\Pi(\mu, \nu)$ is compact and that $\gamma \mapsto K(\gamma) = \int c \, d\gamma$ is continuous, and apply Weierstrass's theorem. We have to choose a notion of convergence for that and we choose to use the weak convergence of probability measures (in duality with $C_b(X \times Y)$, which is the same here as $C(X \times Y)$ or $C_0(X \times Y)$). This gives continuity of K by definition, since $c \in C(X \times Y)$.

As for the compactness, take a sequence $\gamma_n \in \Pi(\mu, \nu)$. They are probability measures, so that their mass is 1, and hence they are bounded in the dual of $C(X \times Y)$. Hence usual weak-* compactness in dual spaces guarantees the existence of a subsequence $\gamma_{n_k} \rightharpoonup \gamma$ converging to a probability γ . We just need to check $\gamma \in \Pi(\mu, \nu)$. This may be done by fixing $\phi \in C(X)$ and using $\int \phi(x) \, d\gamma_{n_k} = \int \phi \, d\mu$ and passing to the limit, which gives $\int \phi(x) \, d\gamma = \int \phi \, d\mu$. This shows $(\pi_x)_\# \gamma = \mu$. The same may be done for π_y . More generally, the image measure through continuous maps preserves weak convergence (and here we use the map $(x, y) \mapsto x$ or $(x, y) \mapsto y$). $\square$ $\square$

Theorem 1.5. *Let X and Y be compact metric spaces, $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$ and $c : X \times Y \rightarrow \mathbb{R} \cup \{+\infty\}$ be lower semi-continuous and bounded from below. Then (KP) admits a solution.*

Proof. Only difference : K is no more continuous, it is l.s.c. for the weak convergence of probabilities. This is a consequence of the following lemma, applied to $f = c$ on the space $X \times Y$. $\square$ $\square$

Lemma 1.6. *If $f : X \rightarrow \mathbb{R} \cup \{+\infty\}$ is a lower semi-continuous function, bounded from below, on a metric space X , then the functional $J : \mathcal{M}_+(X) \rightarrow \mathbb{R} \cup \{+\infty\}$ defined on positive measures on X through $J(\mu) := \int f \, d\mu$ is lower semi-continuous for the weak convergence of measures.*

Proof. Consider a sequence f_k of continuous and bounded functions converging increasingly to f . Then write $J(\mu) = \sup_k J_k(\mu) := \int f_k \, d\mu$ (actually $J_k \leq J$ and $J_k(\mu) \rightarrow J(\mu)$ for every μ by monotone convergence). Every J_k is continuous for the weak convergence, and hence J is l.s.c. as a supremum of continuous functionals. $\square$ $\square$

Box 1.5. – Memo – l.s.c. functions as suprema of Lipschitz functions

Theorem - If f_α is an arbitrary family of lower semi-continuous functions on X , then $f = \sup_\alpha f_\alpha$ (i.e. $f(x) := \sup_\alpha f_\alpha(x)$) is also l.s.c.

Proof - Take $x_n \rightarrow x$ and write $f_\alpha(x) \leq \liminf_n f_\alpha(x_n) \leq \liminf_n f(x_n)$. Then pass to the sup in α and get $f(x) \leq \liminf_n f(x_n)$. It is also possible to check the same fact using epigraphs: indeed, a function is l.s.c. if and only if its epigraph $\{(x, t) : t \geq f(x)\} \subset X \times \mathbb{R}$ is closed, and the epigraph of the sup is the intersection of the epigraphs.

Theorem - Let $f : X \rightarrow \mathbb{R} \cup \{+\infty\}$ be a function bounded from below. Then f is l.s.c. if and only there exists a sequence f_k of k -Lipschitz functions such that for every $x \in X$, $f_k(x)$ converges increasingly to $f(x)$

Proof - One implication is easy, since the functions f_k are continuous, hence lower semi-continuous, and f is the sup of f_k . The other is more delicate. Given f lower semi-continuous and bounded from below, let us define

$$f_k(x) = \inf_y (f(y) + kd(x, y)).$$

These functions are k -Lipschitz continuous since $x \mapsto f(y) + kd(x, y)$ is k -Lipschitz. For fixed x , the sequence $f_k(x)$ is increasing and we have $\inf f \leq f_k(x) \leq f(x)$. We just need to prove that $\ell := \lim_k f_k(x) = \sup_k f_k(x) = f(x)$. Suppose by contradiction $\ell < f(x)$, which implies in particular $\ell < +\infty$. For every k , let us choose a point y_k such that $f(y_k) + kd(y_k, x) < f_k(x) + 1/k$. We get $d(y_k, x) \leq \frac{\ell + 1/k - f(y_k)}{k} \leq \frac{C}{k}$, thanks to the lower bound on f and to $\ell < \infty$. Hence we know $y_k \rightarrow x$. Yet, we have $f_k(x) + 1/k \geq f(y_k)$ and we get $\lim_k f_k(x) \geq \liminf_k f(y_k) \geq f(x)$. This proves $\ell \geq f(x)$. Finally, the functions f_k may be made bounded by taking $f_k \wedge k$.

Theorem 1.7. *Let X and Y be Polish spaces, i.e. complete and separable metric spaces, $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$ and $c : X \times Y \rightarrow [0, +\infty]$ lower semi-continuous. Then Problem (KP) admits a solution.*

Proof. It is now the compactness which is no more evident. We need to use Prokhorov theorem. This means showing that any sequence in $\Pi(\mu, \nu)$ is tight. To do that, fix $\varepsilon > 0$ and find two compact sets $K_X \subset X$ and $K_Y \subset Y$ such that $\mu(X \setminus K_X), \nu(Y \setminus K_Y) < \frac{1}{2}\varepsilon$ (this is possible thanks to the converse implication in Prokhorov Theorem, since a single measure is always tight). Then the set $K_X \times K_Y$ is compact in $X \times Y$ and, for any $\gamma_n \in \Pi(\mu, \nu)$, we have

$$\begin{aligned} \gamma_n((X \times Y) \setminus (K_X \times K_Y)) &\leq \gamma_n((X \setminus K_X) \times Y) + \gamma_n(X \times (Y \setminus K_Y)) \\ &= \mu(X \setminus K_X) + \nu(Y \setminus K_Y) < \varepsilon. \end{aligned}$$

This shows tightness (and hence compactness) of all sequences in $\Pi(\mu, \nu)$. $\square \quad \square$

We add to this section an improvement of the continuity and semi-continuity results above, which could be useful when the cost functions is not continuous.

Lemma 1.8. *Let $\gamma_n, \gamma \in \Pi(\mu, \nu)$ be probabilities on $X \times Y$ and $a : X \rightarrow \tilde{X}$ and $b : Y \rightarrow \tilde{Y}$ be measurable maps valued in two separable metric spaces $\tilde{X}$ and $\tilde{Y}$. Let $c : \tilde{X} \times \tilde{Y} \rightarrow \mathbb{R}_+$ be a continuous function with $c(a, b) \leq f(a) + g(b)$ with f, g continuous and $\int (f \circ a) d\mu, \int (g \circ b) d\nu < +\infty$. Then*

$$\gamma_n \rightharpoonup \gamma \Rightarrow \int_{X \times Y} c(a(x), b(y)) d\gamma_n \rightarrow \int_{X \times Y} c(a(x), b(y)) d\gamma.$$

Proof. We start from the case where c is bounded, say $0 \leq c \leq M$. We can apply the weak version of Lusin's Theorem (see the observations in the next Memo 1.6) to maps valued in $\tilde{X}$ and $\tilde{Y}$. Let us fix $\delta > 0$ and find two compact sets $K_X \subset X, K_Y \subset Y$, with $\mu(X \setminus K_X) < \delta$ and $\nu(Y \setminus K_Y) < \delta$, such that a and b are continuous when restricted to K_X and K_Y , respectively. Let us set $K := K_X \times K_Y \subset X \times Y$, which is a compact set in the product.

We can write

$$\int c(a, b) d\gamma_n \leq \int \mathbb{1}_K c(a, b) d\gamma_n + 2M\delta,$$

and the function $\mathbb{1}_K c(a, b)$ is upper semi-continuous on $X \times Y$ (since it is continuous, positive and bounded on a closed set, and vanishes outside it). This implies

$$\limsup_n \int c(a, b) d\gamma_n \leq \int \mathbb{1}_K c(a, b) d\gamma + 2M\delta \leq \int c(a, b) d\gamma + 2M\delta$$

and, since δ is arbitrary, $\limsup_n \int c(a, b) d\gamma_n \leq \int c(a, b) d\gamma$. This proves upper semi-continuity of the integral functional when c is bounded by M . An analogous computation with $M - c$ instead of c proves lower semi-continuity.

If c is positive but unbounded, just approximate it from below with its truncations $c_M = c \wedge M$, and lower semi-continuity is proven for the integral functional, which will be a sup of lower semi-continuous functionals. By replacing c with the function $\tilde{X} \times \tilde{Y} \ni (\tilde{x}, \tilde{y}) \mapsto f(\tilde{x}) + g(\tilde{y}) - c(\tilde{x}, \tilde{y})$ upper semi-continuity is proven as well. $\square$ $\square$

Box 1.6. – Memo – Lusin's theorem.

A well-known theorem in measure theory states that every measurable function f on a reasonable measure space (X, μ) , is actually continuous on a set K with $\mu(X \setminus K)$ small. This set K can be taken compact. Actually, there can be at least two statements: either we want f to be merely continuous on K , or we want f to coincide on K with a continuous function defined on X . This theorem is usually stated for real-valued functions, but we happen to need it for functions valued in more general spaces. Let us be more precise: take a topological space X endowed with a finite regular measure μ (i.e. any Borel set $A \subset X$ satisfies $\mu(A) = \sup\{\mu(K) : K \subset A, K \text{ compact}\} = \inf\{\mu(B) : B \supset A, B \text{ open}\}$). The arrival space Y will be supposed to be second-countable (i.e. it admits a countable family $(B_i)_i$ of open sets such that any other open set $B \subset Y$ may be expressed as a union of B_i ; for instance, separable metric spaces are second-countable).

Theorem (weak Lusin) - Under the above assumptions on X, Y, μ , if $f : X \rightarrow Y$ is measurable, then for every $\varepsilon > 0$ there exists a compact set $K \subset X$ such that $\mu(X \setminus K) < \varepsilon$ and the restriction of f to K is continuous.

Proof - For every $i \in \mathbb{N}$, set $A_i^+ = f^{-1}(B_i)$ and $A_i^- = f^{-1}(B_i^c)$. Consider compact sets $K_i^\pm \subset A_i^\pm$ such that $\mu(A_i^\pm \setminus K_i^\pm) < \varepsilon 2^{-i}$. Set $K_i = K_i^+ \cup K_i^-$ and $K = \bigcap_i K_i$. For each i we have $\mu(X \setminus K_i) < \varepsilon 2^{1-i}$. By construction, K is compact and $\mu(X \setminus K) < 4\varepsilon$. To prove that f is continuous on K it is sufficient to check that $f^{-1}(B) \cap K$ is relatively open in K for each open set B , and it is enough to check this for $B = B_i$. Equivalently, it is enough to prove that $f^{-1}(B_i^c) \cap K$ is closed, and this is true since it coincides with $K_i^- \cap K$.

Theorem (strong Lusin) - Under the same assumptions on X , if $f : X \rightarrow \mathbb{R}$ is measurable, then for every $\varepsilon > 0$ there exists a compact set $K \subset X$ and a continuous function $g : X \rightarrow \mathbb{R}$ such that $\mu(X \setminus K) < \varepsilon$ and $f = g$ on K .

Proof - First apply weak Lusin's theorem, since $\mathbb{R}$ is second countable. Then we just need to extend $f|_K$ to a continuous function g on the whole X . This is possible since $f|_K$ is uniformly continuous (as a continuous function on a compact set) and hence has a modulus of continuity ω : $|f(x) - f(x')| \leq \omega(d(x, x'))$ (the function ω can be taken sub additive and continuous). Then define $g(x) = \inf\{f(x') + \omega(d(x, x')) : x' \in K\}$. It can be easily checked that g is continuous and coincides with f on K .

Note that this last proof strongly uses the fact that the arrival space is $\mathbb{R}$. It could be adapted to the case of $\mathbb{R}^d$ just by extending component-wise. On the other hand, it is

clear that the strong version of Lusin's Theorem cannot hold for any space Y : just take X connected and Y disconnected. A measurable function $f : X \rightarrow Y$ taking values in two different connected components on two sets of positive measure cannot be approximated by continuous functions in the sense of the strong Lusin's Theorem.

The consequence of all these continuity, semi-continuity, and compactness results is the existence, under very mild assumptions on the cost and the space, of an optimal transport plan γ . Then, if one is interested in the problem of Monge, the question may become “does this minimal γ come from a transport map T ?”. Should the answer to this question be yes, then (MP) would have a solution, which also solves a wider problem, that of minimizing among transport plans. This is the object of the next two sections 1.2 and 1.3. On the other hand, in some cases proving that the optimal transport plan comes from a transport map (or proving that there exists at least one optimal plan coming from a map) is equivalent to proving that (MP) has a solution, since very often the infimum among transport plans and among transport maps is the same. This depends on the presence of atoms (see Sections 1.4 and 1.5).

1.2 Duality

The problem (KP) is a linear optimization under convex constraints, given by linear equalities or inequalities. Hence, an important tool will be duality theory, which is typically used for convex problems. We will find a dual problem (DP) for (KP) and exploit the relations between dual and primal.

The first thing we will do is finding a formal dual problem, by means of an inf-sup exchange.

Let us express the constraint $\gamma \in \Pi(\mu, \nu)$ in the following way : note that, if $\gamma \in \mathcal{M}_+(X \times Y)$, then we have

$$\sup_{\varphi, \psi} \int_X \varphi d\mu + \int_Y \psi d\nu - \int_{X \times Y} (\varphi(x) + \psi(y)) d\gamma = \begin{cases} 0 & \text{if } \gamma \in \Pi(\mu, \nu), \\ +\infty & \text{otherwise,} \end{cases}$$

where the supremum is taken among bounded and continuous functions φ, ψ .

Hence, we can remove the constraints on γ if we add the previous sup, since if they are satisfied nothing has been added and if they are not we get $+\infty$ (which will be avoided by the minimization). Hence we may look at the problem we get:

$$\min_{\gamma} \int_{X \times Y} c d\gamma + \sup_{\varphi, \psi} \int_X \varphi d\mu + \int_Y \psi d\nu - \int_{X \times Y} (\varphi(x) + \psi(y)) d\gamma \quad (1.3)$$

and consider interchanging sup and inf:

$$\sup_{\varphi, \psi} \int_X \varphi d\mu + \int_Y \psi d\nu + \inf_{\gamma} \int_{X \times Y} (c(x, y) - (\varphi(x) + \psi(y))) d\gamma.$$